A low-threat practice-room interface for neurodivergent game learning

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Fast multiplayer games can convert practice into threat when social judgment, sensory load, ambiguous damage, and repeated failure are experienced as one continuous demand. This manuscript reports a single-user design study of league.aolabs.io, a public review room for League of Legends practice. The system does not claim clinical efficacy, population generality, or competitive optimization. It translates a user-provided learning arc into a low-threat review surface: a compact password-free Samira note intake, dense saved-game note cards, parsed game facts from the pasted paragraph, one all-entry Samira takeaway, one direct description per saved note, delete controls for mistaken notes, archived recording review, archived situation notes, public logs, and a downloadable record. The strongest contribution is a design pattern rather than a gameplay result: make the next repetition easier to start by keeping the current capture action first, preserving notes as readable artifacts, and moving recordings, older notes, and other champions into source history until Alan reopens them. The interface treats bots as a practice room, human games as performance, death-with-intent as a valid rep, and recording review as mental practice. This bounded case suggests how personal learning software can support high-repetition play without becoming another source of evaluation pressure.

1 Introduction

Expert performance depends on repeated, structured practice, but practice is only useful when the learner can remain in the task long enough for repetitions to accumulate (*1*). In competitive games, especially player-versus-player games, the immediate source of failure is not only mechanical difficulty. The learner must also manage uncertainty, sensory load, social interpretation, and the felt cost of visible mistakes. Anxiety can impair attentional control (*2*); cognitive load can interfere with learning when too many elements must be processed at once (*3*). A player who is trying to learn a champion, lane, camera, item build, opponent behavior, teammate expectations, and self-regulation at the same time may experience the game as a judgment environment rather than a practice environment.

The system described here began from a simple transfer from music practice: keep the task playable, choose one thing, trust repetitions, and let improvement compound. The same principle was applied to League of Legends. The first rule was not to win lane or outplay an opponent; it was to farm and survive. That rule later compressed into the phrase "next safe minion." As the practice record grew, the problem became less about collecting more advice and more about making advice reviewable without overwhelming the player before queueing.

league.aolabs.io is a public AO Labs page built for that purpose. It is not a Riot Games product, a general League guide, or an official ranked-MMR system. The current home page is a Samira note intake and dense saved-game note cards. Earlier Samira, Caitlyn, and Fizz review-room material remains source history. The design is source-bounded: it uses the user's supplied chat arc, public notes, local recordings, local client reads, and saved Samira note blocks as its corpus, and it avoids claims that would require clinical, population-scale, or competitive outcome evidence.

2 Results

2.1 A long learning arc was reduced to a small review surface

The seed corpus contained a long narrative account of League practice. The arc included lane fear, camera stability, sensory control, champion feel, bot practice, death tolerance, public yapping as pressure regulation, and champion-specific rules for Samira, Caitlyn, and Fizz. The current design result is not a long visible essay, a motivational hero, or a broad champion library. The page opens with a compact two-part Samira work surface: a small width-capped note textarea on the left and one all-entry takeaway on the right, followed by saved game-note cards that are wide enough to avoid skinny text towers. Each saved note shows date, approximate rank, parsed game type/result/KDA/CS/damage details when present, one direct description that extracts the actual note analysis, PDF action, and delete action. That description is not allowed to use assistant preface stems such as "The improvement is" or "The leak is"; it should say the command or fact directly.

This reduction is the main interface result. It makes the review surface compatible with the user's stated constraint: the page should be pretty, clean, non-poster-like, and not overwhelming. The long record remains useful as paper, log, and archive context, but the first screen does not ask the user to process a giant review tray, a generic champion taxonomy, stale situation cards, or raw note rows before saving and reopening the current Samira note.

Table 1. Compression of the source arc into champion-specific review units.

Source burden	Review unit	Interface expression
Enemy lane feels like judgment	Archived situation	Next safe minion remains in the paper record
Damage triggers panic	Mindset situation	Damage is information
Samira has many buttons	Button-order situation	Q before E; W saved for real danger
Low targets bait chase	Chase situation	Low plus reachable, not low plus fog
Champion fit affects repetition	Three champion pages	Samira and Cait recordings; Fizz empty until footage exists

2.2 The interface itself became part of the research record

The public page is not a neutral wrapper around the recordings. It is part of the intervention being studied. For that reason, the paper records the actual interface, not only the backend, manifest, or coaching text. The current live page begins with the shared AO Labs header, a quiet paper strip, one compact Samira note-intake composer, and saved-note PDF cards in a wide dense grid. The earlier champion-card surface used exact user-supplied photographs for Samira, Caitlyn, and Fizz, and those source assets remain in the project archive, but they are no longer part of the default home-page task. This matters because the page is meant to feel like a low-friction intake surface rather than a generic analytics product.

The older top review tile combined a selected champion's rank trend plot and latest synced state into one calm surface. That tile now acts as source history rather than the default interface. The current default card accepts long pasted Samira note blocks without a visible password field, stores them in the same persisted note corpus, and returns one all-entry takeaway plus saved notes with no PDF thumbnails: date, approximate rank, parsed game facts from the paragraph, one direct description that extracts the actual note analysis, one PDF action, and one delete action per note. The input and takeaway are not wrapped in one large blank panel; the note composer stays compact while the saved-note grid uses the page width. The description is rendered through a direct-copy gate so stale stored or generated text cannot wrap the climb-out cue in an improvement preface when the tile should say "Name the climb-out after damage." Older seed PDFs remain reachable by direct route, but they do not fill the home page. The aggregate Samira read and current tips remain available to `/api/samira` and the morning queue. The home page therefore answers one question first: where does the next Samira note go, and what rank-shaped pattern does the whole saved-note corpus imply?

Each recording row is also a designed reading object, but the current visible contract is no longer a coach paragraph. The row now shows a two-column mistake table only: *Time* and *Mistake*, capped at five rows. The table is intentionally narrow because the user asked to reduce overload: it answers only when the mistake happened and what mistake category occurred. The allowed visible categories are red-light E or red-light commit, second fight after value, and chased into bad space. Rank explanation, broad pressure-mode titles, and internal fallback/model wording are not part of the visible review.

Table 2. Current interface elements and the paper obligation they create.

Visible surface	Function in the practice room	Paper record required
Shared AO Labs header and paper strip	Places the tool inside the AO Labs suite and keeps the PDF reachable without adding navigation clutter	Name the public route, suite context, and paper access pattern
Samira note textarea	Captures large pasted practice blocks without making the user enter a review room first	State that the current home page is intake-first and hides recordings, other champions, and public notes by default
Saved game-note reads	Preserve pasted game notes as readable documents while showing one all-entry takeaway, date, rank, parsed game facts, one direct description grounded in the note's actual analysis, one PDF action, and one delete action per note	Reject assistant preface stems in card copy and describe the visible saved-note surface, archived PDF routes, delete boundary, and source-bounded rank boundary
Password-free save state	Allows persisted Samira note writes and deletes without a visible token field	Describe the saved/error feedback state without implying a password step
`/api/samira` read	Feeds rank estimate, saved-note PDF metadata, and morning-email tips from notes plus source history	Describe the source-bounded rank/tip contract without showing recordings as a home-page dashboard
Recording archive and manifest	Preserve timestamped review evidence for later analysis	State that recordings are source/archive material, not the default visible page
Older champion-card assets	Preserve exact supplied photos and prior fit comparisons	State that other champions are retained as source history unless Alan reopens that surface

2.3 Recordings moved behind the Samira intake

The site now organizes the visible page by Samira note intake and saved-game note PDFs rather than by abstract lesson cards, champion choice, recording review, or an archive grid. Situation notes and recording rows still exist because they explain the origin of the rules and preserve evidence, but they are no longer loaded into the default home page. This format was narrowed because the user needed less cognitive load than a giant rule wall, a stat audit, a champion picker, a repeated note archive, or a paragraph coaching taxonomy.

Table 3. Example Samira rules retained in the archived situation record.

Situation prompt	Combined response
The fight feels urgent and E wants to become the first button.	Start with Q or an auto so the fight gives information before the dash commits the body.
The enemy has CC that is not recognized yet.	Stay outside on Q-only, watch one danger spell, use W or sidestep for visible projectiles, and E only after the scary spell is gone.
A low-health enemy runs toward fog or teammates.	Stop at the edge; low only matters when the target is also reachable and cheap to kill.
W feels necessary immediately after dashing in.	Hold W for the real spell; it is a parry for visible danger, not armor against being close.
R works and the excitement makes staying feel automatic.	Treat R as the reward, then leave unless the next target is already low and reachable.
E resets after a takedown.	Read the reset as a new check, not an instruction to dash into tank, tower, fog, or fresh crowd control.
One hit lands and feels like proof that the enemy is competent.	Treat damage as information, step back, stabilize the camera, and choose the next action.
The player is near a team fight, R is not ready, and waiting there costs HP.	Move to the edge, build style from safety, and enter only when S rank exists and the fight is already committed.
The player tunnels on one low target.	Check target, danger, target, danger; stop chasing if the chaser becomes the low target.

2.4 Other champions became source history

The champion comparison material is no longer a visible picker on the home page. It remains source history. Samira remains the active intake target because she provides a high-reward dash-reset-R loop, but she also creates the central panic trap of using E as an anxiety button. Caitlyn remains the comfort-sniper reference for range, farming, and unavailable positioning. Fizz remains the comparison reference because his button shape makes the desired loop clear: mark, enter, stab, dodge, leave.

The older champion-fit material remains useful as source history, but it is not allowed to compete with the current practice page. The public page uses the Samira intake as the selection mechanism and lets recording or champion-specific feedback stay hidden until Alan explicitly reopens that surface. This avoids fake specificity and preserves the rule that no footage means no feedback.

2.5 Raw recordings converted carry games into timestamped improvement lanes

On May 18–25, 2026, the project added a generated recordings manifest from the local League of Legends Highlights folder and the live capture publisher. The current manifest contains 55 recordings across 43 matches and includes short highlight clips plus longer review recordings. Champion detection is bounded to source evidence: older rows remain Samira when sampled frames show Samira in the team list or center the Soul Fighter Samira model and nameplate, while the newest Cait row is source-confirmed from the local client champion id and visible Cait review footage. No champion label is treated as source-confirmed without that kind of evidence.

The recording review does not attempt to become a full coaching dashboard. When reopened, recording rows use a two-column mistake table instead of a narrative paragraph: *Time* gives the clickable game-clock anchor, and *Mistake* gives one short allowed category plus the concrete wrong action and replacement action when visible. Rows are capped at five mistakes so the review answers the immediate question without forcing the user to parse rank explanations, paragraph color semantics, or internal audit language. The performance-rank estimate still exists in the manifest and API source graph, using Riot's current tier structure and queue distinctions (*4*, *5*), but it is no longer a home-page surface. Co-op vs. AI rows remain lower-confidence ranked-habit evidence, while ranked/PvP full games remain the stronger transfer evidence. If the detailed model pass fails or returns no usable JSON, the sync builds a conservative table from verified clock frames and hides internal fallback wording from the visible review.

The May 24 dense-review update keeps that example-review contract but reduces visible repetition and whitespace. Every new current-standard full-game row carries a source-backed victory or defeat label when the League Client match history provides it, uses one sharp mistake title, and starts with one natural key timestamp sentence that names the exact visible state, the wrong click, and the correct next click. Red text is reserved for the leak stated once, green text for one concrete habit to keep, and a distinct warm rose-pink text for one ranked-game *Rep:* drill; the paragraph no longer bolds those spans, so color rather than weight carries the skimmable meaning. The pink drill is now selected from the recording-specific mistake family instead of reused as a universal checklist: death-heavy games drill low-HP exits and no re-entry, death-heavy lane games drill the exact no-dash-under-tower support-and-wave condition, high-kill losses drill cashing out after the first won exchange, grouped base-push wins drill structure–blocker–wave–exit click order, side-farm leaks drill threatened-turret and ally-death checks, and cleaner wins drill the one remaining exit branch while preserving the improvement signal. Neutral text supplies only the timestamped state, consequence, and K/D/A–CS context needed to support those colored claims. At desktop width the recording row uses a balanced text/video split with larger coaching type, so the paragraph and replay are peers rather than a small note beside a dominant video. Rows are rejected when they fall back to generic review-frame language, label lists such as “mistake category” and “correct next click,” repeated tower–wave–objective–front checklists, a pink drill that does not match the game-specific mistake type, vague “permanent thing” drill wording, or abstract branch terms that do not become an if-this-then-that click rule.

The May 25 forensic-performance update adds a stricter rank and timestamp contract. A row now reports one *Approx performance rank* for that game, never a range, with a one-sentence reason tying together CS/min, K/D/A and death count, fight-entry legality, exit or re-entry quality, and whether pressure became objective, wave, recall, group, or structure value. The key timestamp must separate the play into phases in natural prose: legal or illegal entry, value created or not, state flip, first bad next click, and correct next click. Objective-fight reviews are calibrated so the first entry is not called wrong just because the final result was bad; if dragon or Baron is visible, allies are committed, and enemies are already engaged, the review can mark the entry partly legal and critique the second fight after value instead. The pink drill remains game-specific: lane deaths drill first safe exit and no illegal E, objective fights drill dragon/wave/recall/group after first value, fed losses drill the first won fight into objective/tower/wave/recall, grouped base pushes drill structure/blocker/wave/exit, and jungle-fight exits drill turret, mid wave, reset, or regroup after the first value window. A May 26 micro-feedback correction adds the same specificity requirement to Samira fight rows, not only the early-lane side pass: if the timestamp is a lane trade or mid-fight re-entry, the review must say whether the first entry or damage window is playable, identify the first auto/Q or burst window, name the back-click or reset-spacing point, and state whether the next E or forward click is legal or just a chase. The same branch gate rejects recycled objective, farm, pressure-mode, or macro-only phrasing when the title, timestamp, red leak, and pink rep do not describe the same actual play type.

The May 27 Samira update narrows future reviews to a single habit until the user changes the standard: green-light discipline for `E` toward champions. A Samira row must first ask whether all three fight conditions are true: `W` ready, HP above half, and an ally on screen or close enough to fight with her. If any condition fails, the moment is red light: `W` gray means no `E`, below-half HP means no `E`, and no nearby ally means no `E`. The review must find the key timestamp where this rule breaks, call the moment red light or green light, state which condition failed, name the wrong input, and translate the correction into a literal input such as Q/auto while backing up, kite, farm, wait, or reset spacing. Macro language about rotations, broad cash-out, or objective conversion is demoted unless it directly follows from a green-light violation. The pink *Rep:* line is therefore no longer allowed to drift into a generic map drill for Samira; it must preserve the next-game cue "W, HP, ally" and the operational rule that red-light Samira farms, kites, or waits while green-light Samira can fight.

The latest May 27 overload-reduction update removes paragraph reviews from the visible recording cards. Future public reviews are mistake tables only, with columns *Time* and *Mistake*, five rows max. The table vocabulary is deliberately small: red-light E or red-light commit, second fight after value, and chased into bad space. The row text must be short and concrete, for example naming which green-light condition failed, that a second fight happened after value, or that a chase moved into tower, brush, river, jungle, or enemy side. Rank explanations, broad "pressure mode" labels, and internal fallback/model wording are rejected at the manifest audit layer.

The later May 25 rank-audit pass recalibrated every full-review row in the 55-recording manifest rather than only the newest sample. The audit rejects Gold-or-higher grades without Gold-level CS/min, death count, queue pressure, and conversion evidence; caps Co-op vs. AI rows below transferable high-rank claims; keeps low-impact or death-heavy games in Iron unless carry-level damage and farm justify a Bronze floor; preserves known Ranked Solo queue metadata from earlier local-client evidence when current logs no longer expose the queue id; and requires each visible rank reason to mention CS/min, deaths or K/D/A, entry or fight context, and exit or value conversion in one sentence. This calibration changed the rank trend by removing inflated Gold/Silver reads and by treating wins, losses, bot games, and high-kill losses as performance evidence rather than outcome labels.

2.6 The public log made mindset part of the artifact

The public log starts with source entries for the seed arc import, the practice-room frame, the current Samira law, and later user-supplied Samira proof-game notes. The Samira intake stores longer pasted paragraphs in that same persisted note corpus and exposes them through a compact read endpoint for review, note PDFs, per-note rank reads, and morning-email tips. This is intentionally public. The design choice is that thoughts, panic notes, and mindset rules are not private by default; they are part of the learning artifact. The only excluded material is unrelated sensitive information such as credentials, access tokens, or private third-party identifiers.

3 Discussion

The case suggests that the useful interface for some high-repetition learners is not less information in the abstract, but the right default object. A long learning conversation can contain many correct rules while still being hard to use before a game. The current home page therefore removes the champion picker, recording review room, archived situations, public note stream, older note archive, decorative PDF preview, and oversized enclosing intake panel from the default surface. The page now opens as one compact password-free Samira note textarea, save state, one all-entry takeaway, and dense saved-note reads showing date, rank, parsed game facts, one direct description grounded in the note's actual analysis, PDF action, and delete action. The note composer stays width-capped while the saved-note grid can use the wider page for many games, with card widths chosen to avoid narrow paragraph towers. The saved-note description is direct-action copy, not category-prefaced coaching prose. Older PDFs, recordings, and distilled Samira tips remain available to the API and morning queue. That is consistent with cognitive-load theory: reducing simultaneous elements can make practice more tractable (*3*). It is also compatible with mental practice research, because the stored notes can still become concrete game cues without making the capture surface feel like another review assignment (*6*).

The work also reframes death and failure. The site does not encourage intentional feeding or fake failure. It uses the rule "choose the fight; death is allowed; winning is intended." That distinction matters. It keeps agency with the learner while preserving the real objective of the game. It also separates practice failure from collapse: a failed engage that tested damage, W timing, or exit timing is a rep; walking in only to die is not.

Several limits are important. First, this is a single-user artifact, not a clinical intervention. The user self-described ADHD/autism traits and supplied a practice narrative, but the site does not diagnose, treat, or generalize to neurodivergent players as a class. Second, the current evidence is artifact-level: source text, distilled rules, local recordings, local League Client match-history reads, a rendered page, and the paper itself. It does not include a controlled comparison or a ranked outcome. Third, the system deliberately avoids public Riot API import in version 1. That keeps the artifact focused on review and self-regulation rather than turning it into a general stat tracker.

4 Materials and Methods

Source corpus

The seed corpus was the user-provided League learning arc pasted into the implementation thread on May 17, 2026. It described the user's own League practice, including early Caitlyn farming goals, Samira drills, bot-ladder practice, camera and sensory adjustments, death exposure, and champion-fit comparisons. Later May 17 and May 18 prompts added Samira crowd-control, W-blocking, poke-lane, bait-recognition, team-fight, bad-team, normal-game, and proof-game refinements. A May 18 local recording folder then supplied 15 files across matches `NA1-5563247854`, `NA1-5563301586`, and `NA1-5563352800`. Local replay files supplied the match-time anchor, League client logs supplied queue id `880`, Riot's published queue table identifies `880` as Co-op vs. AI Beginner Bot games,(*7*) and the local League Client match-history endpoint supplied K/D/A, CS, and game duration when the client was available. The implementation treated these texts, recordings, replay files, local logs, and local client reads as the current source of truth for the public situation notes and recording review.

Interface requirements

The page was constrained by user instructions gathered during planning and revision: one page only, clean and pretty, same AO Labs vibe, no poster hero, no dense dashboard, no redundant helper labels, no overexplaining, no bold card hierarchy when Alan rejects it, one quiet paper strip, and no visible recording or other-champion material when the current job is Samira note intake. Earlier paragraph reviews used green, red, and pink coach clauses, and a later review mode replaced them with a mistake table. Those systems are now source/archive layers, not the default home page. The current visible interface uses the shared AO Labs header, warm off-white palette, dark ink text, restrained borders, one visible paper action, one compact password-free Samira note input, one all-entry takeaway, and dense saved-note reads showing date, rank, parsed game facts, one direct description grounded in the note's actual analysis, PDF action, and delete action. It avoids the rejected pattern of a large mostly empty intake card above cramped narrow note tiles. Card copy removes both role-prefix labels and colonless preface stems; the visible sentence should read like "Name the climb-out after damage," not an improvement-wrapper around the same command. The input width is independent from the saved-note grid width. This interface description is treated as part of manuscript accuracy: when the page changes in a way that changes what Alan sees, how evidence is displayed, how queue state is communicated, or how coaching intent is encoded, the paper must be checked or updated in the same work cycle.

Rule extraction

Rules were extracted by finding repeated gameplay or mindset triggers in the source corpus and translating each into a one-sentence situation prompt plus one combined response paragraph. The extraction favored recognition over slogan compression: the prompt had to name the exact stressful moment, and the response had to combine action, reason, and practice rep without separate "do," "avoid," or "drill" labels. For example, the Samira section retained the underlying rules "Q first; earn the E," "W is parry, not armor," and "R is reward, not start," but the public card now starts from the situation that makes the rule necessary. The death section retained "choose the fight; death is allowed; winning is intended." The bot-practice section retained the practice ladder and the current-mode cue.

Public artifact implementation

The website is served by a small Node application on Railway. It serves the hand-authored page, generated recording manifest, exact champion-card photos, poster frames, and paper artifacts from the public directory and exposes `/api/logs` for public note reads and writes plus `/api/samira`, `/api/samira/notes`, `/api/samira/notes/:id`, and `/api/samira/notes/:id.pdf` for the Samira note/rank/tip, delete, and PDF read. The home page now initializes only the Samira note form and deduplicated saved-game note PDF cards; it does not hydrate the recording manifest, recording heartbeat, champion picker, selected-champion panel, archived situations, public note stream, or older seed note archive. Those sources remain available to backend reads and future review work. Local recording media is served with byte-range responses so timestamp controls can seek directly into a video row when that archive is reopened. Large recording videos are kept out of the Railway upload payload; when a local video file is not present in the deployment image, the server redirects that recording request to a GitHub-hosted media copy for the same repository path. A local sync script scans the source Highlights folder, reads replay and League client logs for match and queue metadata, reads local League Client match history for champion id, K/D/A, CS, and game duration when available, copies or compresses recordings into deployable assets, extracts poster frames with `ffmpeg`, computes the research-calibrated rank trend, and refreshes the manifest. The current manifest review gate requires a mistake table for full automatic reviews: one to five rows, each with a verified game-clock time and one allowed mistake category. Samira rows audit green-light discipline first: if `W` readiness, above-half HP, or nearby ally presence is absent or not visibly proved, a forward champion commit is red light and the row must name the non-`E` input. The normal macro capture stays at 2 FPS so whole-game review remains low-lag and cheap, while the first 12 minutes can still be captured at a bounded higher frame rate for a separate early-lane micro pass. On Railway, persistent storage is provided through the configured volume path; Samira note save and delete are password-free public operations in this version. This keeps public notes browser-independent without adding a fake local-only note form or claiming public Riot API integration that is not present in version 1.

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Author contributions

Alan N. Pham supplied the practice corpus, product constraints, and acceptance criteria. The manuscript and site were assembled from those source materials in the AO Labs workspace.

Competing interests

The author owns and operates AO Labs. The site is a personal public artifact and is not endorsed by Riot Games.

Data availability

The first public data source is the League practice arc supplied in the implementation thread and the Samira note intake exposed through league.aolabs.io. The current public home page exposes the Samira note form and deduplicated saved-game note PDF cards. The API still exposes generated recording manifest data, poster frames, local queue metadata, local client champion id, K/D/A and CS when available, per-recording mistake tables, research-calibrated rank-equivalent macro estimates, the Samira note/rank/tip read, public note PDFs, compact post-game queue status, and timestamp anchors that seek into playable videos when archive views are reopened. The manifest also exposes a rank-calibration block with the research sources, high-confidence requirements, and confidence blockers that separate ranked/PvP rows from bot-practice rows. Public Riot API match history is not imported in this version.

Code availability

The implementation is intended for the `nalalalan/league-app` repository, with the public site, PDF, manuscript source, and log API served at league.aolabs.io.

Additional information

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